

Experiment 3

1. Aim

To design and simulate a **4-Bit Ripple Carry Adder** in **Logisim Evolution** by cascading four Full Adder circuits, and to verify that the circuit correctly computes the binary sum of two 4-bit numbers along with a carry-out for all relevant input combinations.

2. Theory

A **Ripple Carry Adder (RCA)** is a combinational circuit that adds two multi-bit binary numbers by chaining Full Adders in series. In a **4-bit RCA**, four Full Adders (FA0–FA3) are connected such that the carry-out of each stage feeds into the carry-in of the next stage — the carry 'ripples' from the least significant bit to the most significant bit.

Each Full Adder computes:

- $\text{Sum}_i = A_i \oplus B_i \oplus C_i$
- $C_{i+1} = (A_i \cdot B_i) + (B_i \cdot C_i) + (A_i \cdot C_i)$

The final 4-bit output is S3 S2 S1 S0, with C_{out} from FA3 serving as the overflow/carry bit. The main limitation of the RCA is propagation delay — the total delay grows linearly with the number of bits since each stage must wait for the previous carry.

Sample Input–Output Combinations:

A3A2A1A0	B3B2B1B0	C_{in}	Sum (S3S2S1S0)	C_{out}
0000	0000	0	0000	0
0001	0001	0	0010	0
0011	0101	0	1000	0
0111	0111	0	1110	0
1001	0110	0	1111	0
1111	0001	0	0000	1
1111	1111	0	1110	1
1111	1111	1	1111	1

3. Procedure

1. Open **Logisim Evolution** and create a new project file.
2. Build a **Full Adder sub-circuit**: create a new circuit named 'FullAdder' using 2 XOR gates, 3 AND gates, and 1 OR gate. Define input pins A, B, C_{in} and output pins Sum and C_{out} .
3. In the main circuit, place **four instances** of the Full Adder sub-circuit (FA0, FA1, FA2, FA3) on the canvas.

4. Add 8 input pins for the two 4-bit numbers: **A0–A3** and **B0–B3**. Add one input pin for the initial C_{in} (tied to 0 for normal addition).
5. Connect the corresponding A and B bits to each Full Adder: A0, B0 → FA0; A1, B1 → FA1; A2, B2 → FA2; A3, B3 → FA3.
6. Wire the **carry chain**: connect C_{out} of FA0 → C_{in} of FA1, FA1 → FA2, FA2 → FA3.
7. Add output pins **S0–S3** connected to the Sum outputs of FA0–FA3 respectively. Add a final output pin C_{out} from FA3. Connect outputs to a hex display.
8. Use the **Poke tool** to apply various 4-bit input combinations and verify the Sum and Carry outputs against expected binary addition results.
9. Take a screenshot of the completed and verified circuit.

4. Circuit Design / Implementation

The 4-Bit Ripple Carry Adder was implemented in **Logisim Evolution** as follows:

- **4 Full Adder blocks** — each built from 2 XOR, 3 AND, and 1 OR gate, implemented as a reusable sub-circuit.
- **Carry chain** — C_{out} of each stage is directly wired to C_{in} of the next, shown in red to distinguish carry wires from data wires (green).
- **8 input pins** (A3–A0, B3–B0) and **5 output pins** (S3–S0, C_{out}) with a hex display for real-time result monitoring.

Green wires carry data signals; red wires carry the ripple-carry chain. The four Full Adder stages are arranged vertically, with carry propagating downward from FA0 (LSB) to FA3 (MSB).

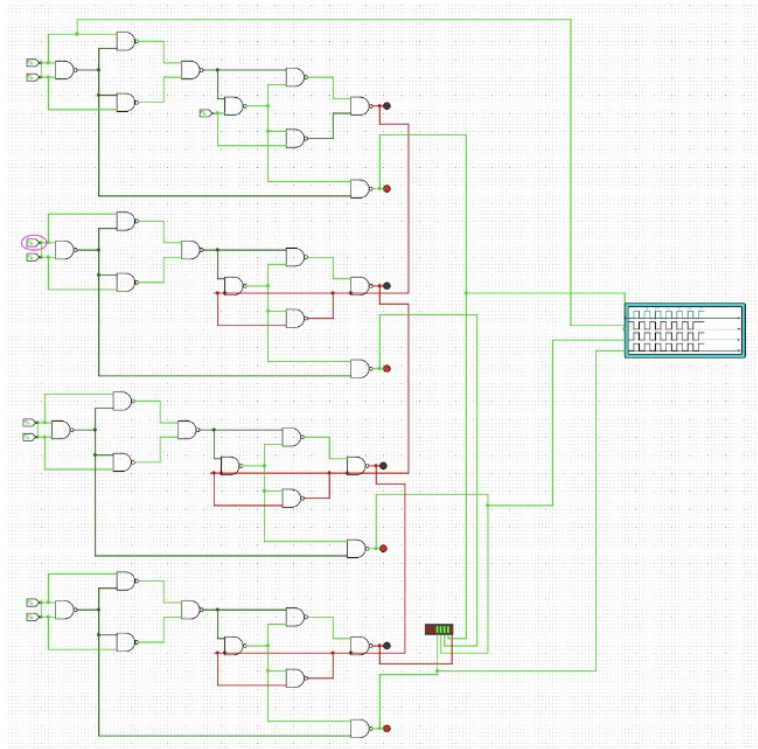


Figure 1: 4-Bit Ripple Carry Adder in Logisim Evolution — four Full Adder stages (FA0–FA3) with carry chain (red) rippling from LSB to MSB. Outputs connected to a hex display for verification.

5. Result

The 4-Bit Ripple Carry Adder was successfully designed and simulated in Logisim Evolution. All tested input combinations produced the correct 4-bit Sum and Carry-out values, matching the expected binary addition results. The carry signal was observed to propagate correctly through each Full Adder stage, confirming the ripple-carry mechanism.

6. Conclusion

This experiment demonstrated the construction of a **4-Bit Ripple Carry Adder** by cascading Full Adder circuits in Logisim Evolution. The design validated how carry propagation links individual 1-bit adders into a multi-bit arithmetic unit. The RCA serves as a fundamental component in ALUs and binary arithmetic systems, and understanding its delay characteristics lays the groundwork for studying faster adder architectures such as the Carry Lookahead Adder.